TITLE: Mapping of extracellular RNA from human biofluids reveals >17 million genomic loci covering >18% of the human genome that express extracellular RNA

RUNNING TITLE: Genome-wide map of human extracellular RNA

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ABSTRACT

Integrative analysis of 6,532 extracellular RNA (exRNA) profiles from human serum, plasma, cerebrospinal fluid, urine, and saliva across 60 independent datasets in the exRNA Atlas reveals >17 million genomic loci, covering >18% of the human genome that express exRNA. A core set of exRNAs covering ~0.8% of the genome are detectable across all biofluids while others show variable presence and abundance between biofluids. To support future exRNA research, we develop a comprehensive resource that assigns unique identifiers (exGLid) to each exRNA locus. We annotate the exGL loci with analysis results and distinct classes of overlapping transcribed (GENCODE) and regulatory (ENCODE) loci. We make this information accessible via user interfaces, APIs, and visualization tools. Taken together, our results demonstrate the potential of this first-of-its-kind exRNA map to provide valuable insights into exRNA biology and catalyze future exRNA research.

INTRODUCTION

The study of extracellular RNA (exRNA) communication was initiated by the seminal discoveries of intercellular transfer of mRNA and miRNA molecules and their effects on gene expression in target cells. The extracellular mRNA transcripts were shown to be translated into protein in target cells (Ratajczak et al. 2006; Valadi et al. 2007; Skog et al. 2008), and the intercellular transfer of miRNAs diminished translation of target mRNAs upon miRNA uptake by target cells (Valadi et al. 2007; Pegtel et al. 2010; Montecalvo et al. 2012). These pioneering discoveries opened the question about the full extent of exRNA-mediated intercellular gene regulation. The advent of next-generation sequencing technology and the availability of the human reference genome provided an opportunity to generate foundational knowledge for research into that question. The NIH Common Fund exRNA Communication Consortium (ERCC) (Das et al. 2019; Ainsztein et al. 2015; Mateescu et al. 2022; Amolegbe et al. 2025) was established in 2013 to harness emerging technologies and informatics to accelerate progress in the nascent exRNA biology field. By aggregating exRNA profiles from

numerous small RNA-seq (smRNA-seq) profiling studies, ERCC produced early versions of the exRNA Atlas (Murillo et al. 2019; Subramanian et al. 2015; LaPlante et al. 2023), which consists of maps of exRNAs and their vesicular and non-vesicular carriers in human biofluids. We here report on the third version of the ERCC exRNA Atlas which includes 6532 exRNA profiles generated over the ten years of the consortium. This version provides the first comprehensive map of exRNA loci in the human genome.

Previous versions of the exRNA Atlas focused on distinct aspects of exRNA biology. The first version mapped exRNA to its vesicular (exosomes and other extracellular vesicles) and non-vesicular (lipoproteins, RNPs, and other specific particle classes) carriers in human biofluids (Murillo et al. 2019), and the second version deepened our understanding of exRNA carriers by identifying extracellular RNA-binding proteins (exRBPs) that associate with exRNA fragments in the extracellular space (LaPlante et al. 2023). One key limitation of those early maps was that the exRNA expression was quantitatively ascertained using GENCODE and ENCODE/ENCORE (Van Nostrand et al. 2020) genome annotations. While being useful as initial stepping stones toward understanding exRNA biology, these annotations heavily constrained the insights that could be gained about the role of exRNA in human biology.

The most glaring gap in exRNA biology was the lack of information about the loci that are transcribed into exRNA (Goh et al. 2022). With the completion of the ERCC and aggregation of the uniformly processed exRNA profiles in the Atlas, we here address this gap for the first time. We provide the first map of extracellular Genomic Loci (exGLs) transcribed and exported as extracellular RNA into human blood (plasma and serum), urine, cerebrospinal fluid (CSF), and saliva.

These five biofluids were selected because they represent the largest number of samples in the exRNA Atlas, comprising 6,532 smRNA-seq profiles from 60 datasets. For each of those biofluids, we identify biofluid-specific exGLs *de novo*, independent of previous genome annotations.

To support standardized referencing of these exGLs, we developed the exGL identification service, which assigns unique identifiers (exGLids) to each exGL. exGLids and associated annotations, such as expression depth, gene or regulatory element overlap, and biofluid distribution, are available for full download across all five biofluids and can also be accessed programmatically via API endpoints and for visualization via a UCSC track hub. To facilitate exRNA biology research, we provide software tools and APIs for exGL query and analysis by gene, genomic region, or biofluid.

We here present a summary of the exGL map within the exRNA Atlas resource and insights gained from its analyses, including the exRNA biotype diversity, abundance patterns, and analyses of negative selection.

RESULTS

Section 1: Plasma, serum, urine, CSF, and saliva exGLs are typically short and in aggregate cover 18.35% of the human genome.

Our analysis focused on the five biofluids with the largest number of small exRNA-profiled samples and datasets within the exRNA Atlas: plasma (3357 samples, 42 datasets), serum (1817 samples, 15 datasets), CSF (647 samples, 8 datasets), urine (363 samples, 6 datasets), and saliva (348 samples, 5 datasets). In this work, a dataset refers to a collection of samples profiled by exRNA sequencing and submitted as a unit (typically from one submitter). Each dataset is tracked under a single exRNA Atlas identifier. The per-dataset sample counts, donor-condition breakdowns, RNA isolation kits, and library construction kits are provided in **Supplementary Table S1**. We represented the small exRNA-seq profiles of samples as base pair-resolution read density maps in bedGraph format. For each sample, we identified loci contiguously covered by exRNA reads and filtered out loci <18 bp to avoid artifacts from non-specific read mapping (**Figure 1A**). The 18 bp threshold is consistent with existing small RNA-seq processing standards including the default minimum read length in exceRpt (Rozowsky et al. 2019) and the default 18 nt genome-mapping seed used by miRMaster (Fehlmann et al. 2021)

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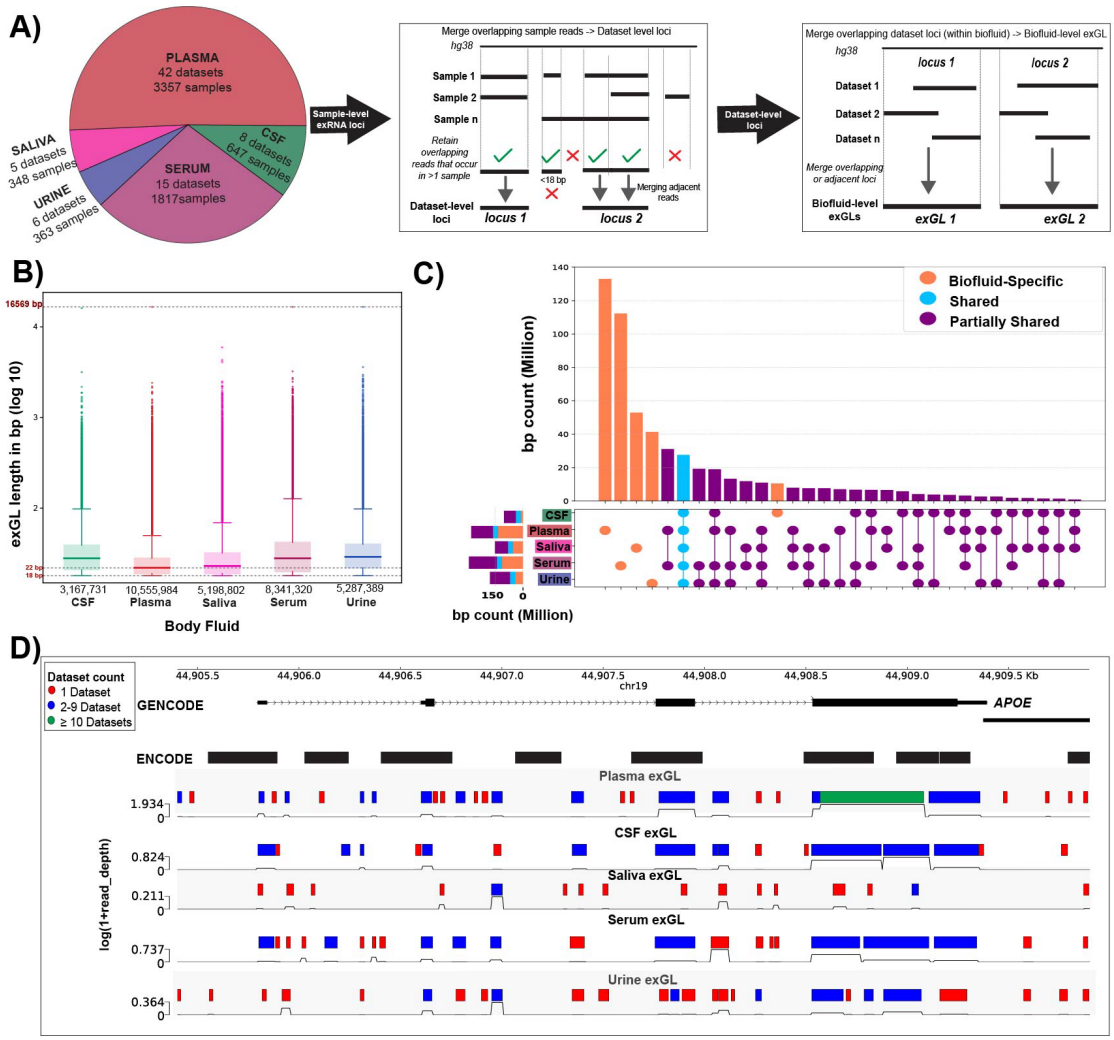


Figure 1: Identifying, annotating, and characterizing exGLs across biofluids. *(A) Schematic overview of the workflow for defining exGLs. Overlapping regions of exRNA reads are first merged (intersection-based merging) across samples of each dataset to retain loci present in more than one sample (≥18 bp). Individual dataset loci are then merged (union-based merging) across multiple datasets within a biofluid to define biofluid-specific exGLs. (B) exGL size distributions across five biofluids (CSF, Plasma, Serum, Saliva, Urine). Significant differences in exGL length distributions are observed between biofluids ($p < 2.22e-16$, Wilcoxon rank-sum test). Numbers below boxplot indicate the total number of exGLs in each biofluid. (C) Overlap of exGLs across biofluids. The bars (horizontal) represent the number of base pairs (log scale) corresponding to each biofluid-specific (orange), shared (light blue), or partially shared (purple) exGL intersection. (D) Genomic visualization of exGLs from each biofluid in the APOE gene region as an example. The normalized read depth of the exGLs are shown in the line plot. Colors of the exGL correspond to the number of datasets contributing to the exGLs (red: 1 dataset, blue: 2-9 datasets, green: 10+ datasets).*

The exRNA genomic loci (exGLs) were constructed separately for each biofluid. For each dataset, exRNA loci present within at least two samples were merged to create dataset-level loci. Biofluid-specific exGLs were created by merging all dataset-level loci (**Figure 1A**, **Supplementary Table S2**). For plasma - the biofluid with the largest number of datasets (42 plasma datasets) - the process identified 10.55 M exGLs covering ~9% of the genome (277 Mbps).

Most examined exGLs across biofluids were short, with some notable exceptions. The mean and median plasma exGL size was 26.3 bp and 22.0 bp, respectively, with 99% of exGLs between 18 and 200 bp. The largest plasma exGL (16,569 bp) corresponded to mitochondrial chromosome loci that merged into a single exGL. The size distribution was similar for other biofluids, with serum having the largest mean exGL size (34.7 bp), followed by urine (33.4 bp), CSF (31.9 bp), and saliva (28.9 bp) (**Figure 1B**). While the distribution of exGL sizes across biofluids was similar, the smallest mean exGL size (26.3 bp) in plasma was surprising given that plasma was represented by the largest number of datasets within the exRNA Atlas. To investigate this further, we examined the trend of exGL number versus dataset size by progressively adding plasma datasets to create exGLs. As the number of datasets increased, merging of overlapping loci from different datasets caused some exGL to combine and become longer. However, the mean exGL size exhibited a local peak pattern, initially rising to a peak (~28 bp) before stabilizing at ~26 bp (**Supplementary Figures 1A and 1B**). These findings suggest that the addition of new datasets does not directly lead to an increase in exGL size (as the addition may result in new, shorter exGLs being discovered, thus reducing the average size). The increase in exGL size with dataset accumulation may also depend on the order in which datasets accumulate and interactions between biofluid-specific or dataset-specific properties and experimental variables such as RNA extraction or library preparation methods (Srinivasan et al. 2019).

We further examined reproducible detection of exGLs from different plasma datasets. A core set of ~127K exGLs (~1.2% of the genome) were consistently detected in 10 or more plasma datasets despite experimental and biological variability, suggesting that the loci covering at

least 1% of the genome have the potential to be reproducibly detected. While the core set was sizable (**Supplementary Figure S1C, Supplementary Table S3**), each plasma dataset led to a significant number of novel exGLs (**Supplementary Figure S1A**). Longer exGLs were more frequently observed across datasets, which may in part reflect the higher probability of their sampling by random reads. A similar trend is observed across all biofluids (**Supplementary Figures S1D-1H**)**.** For example, in the *APOE* region, longer exGLs were more frequently observed not only across datasets within a biofluid, but also across biofluids (**Figure 1D**). Taken together, these patterns suggest that the depth of sequencing of current smRNA-seq profiling experiments may not reflect the remarkable complexity of exRNAs. At current sequencing depths, such experiments likely detect only the more abundant and readily sampled exRNAs, leaving a substantial portion of the exRNA landscape unexplored.

We next projected exGLs from different biofluids onto the genome to determine the degree of overlap of exGLs from different biofluids. When merged, the exGLs from all the five biofluids covered ~17 million contiguous loci covering 18.35% of the genome. Only a fraction of exGLs (4.4%) shared precisely the same coordinates across two or more biofluids, while many more overlapped among biofluids. To evaluate concordance between biofluids at the basepair level of resolution, we classified each exGL basepair as biofluid-specific (unique to a biofluid), shared (present in all five biofluids), or partially shared (present in 2-4 biofluids) (**Figure 1C***)*. Approximately 27 Mbps were shared across all biofluids, suggesting that > 0.8% of the genome can be consistently detected as exRNA across the biofluids examined. In contrast, > 11% of the genome (350 Mbp) was specifically detected within just one of the five biofluids.

The *APOE* gene region provides an illustrative example of coverage by exGLs from multiple biofluids, with some loci being biofluid-specific while others being covered by exGLs from multiple biofluids (**Figure 1D**). We tested whether cross-biofluid concordance exceeds the degree of overlap expected by chance. Toward this end, for each biofluid we estimated the probability of a basepair being within an exGL. Using these probabilities, using a null multinomial model, we calculated expected numbers of base pairs shared by 0-5 biofluids.

The number of base pairs included in exGLs across 0, 4 and 5 biofluids was larger than expected, suggesting concordance across biofluids. This enrichment was not specific to *APOE*. We repeated the analysis on 1,000 randomly sampled genes and found the same pattern (**Supplementary Figure S2A-2B**). We next asked which patterns also occurred within the subset of exGLs that overlap eCLIP sites (Van Nostrand et al. 2020) of RNA binding proteins (RBP). We focussed on the 34 RBPs previously experimentally validated to bind to plasma exRNA (LaPlante et al. 2023). The number of base pairs included in exGLs across 0, 4 and 5 biofluids was again larger than expected (**Supplementary Figure S2C**), suggesting concordant basepair coverage across biofluids for RBP-associated exRNA.

Section 2: The abundance of plasma exGLs is highly unevenly distributed, with a highly abundant subset and a long tail of rare exGLs

We next explored the variation in exGL abundance, with a focus on plasma because it had the highest sample count (n=3357 samples) among all biofluids. Sample read depth varied across datasets, individual samples, and loci. Therefore, the variation in the abundance for a specific exGL could not be conclusively interpreted as being due to the biological variation in the abundance of specific RNA fragments in specific biofluid samples. To address this variability, we applied averaging over many exGLs that showed similar abundance. Toward this end, we divided exGLs by abundance (ascertained by read depth) and for each abundance class quantified attributes such as percentages of each exRNA biotype and the probability of detection. We did not impose a hard abundance cutoff for exGL inclusion, with the rationale that because some exRNA datasets may be under-sequenced, a fixed threshold might discard true loci that are under-sampled due to relatively low sequencing coverage.

We calculated exGL read depth through the following two-step process: Step 1 (Estimate dataset-level read depth): Loci across samples within a dataset were merged to create dataset-specific loci; normalized read depth was then averaged across merged loci. Step 2 (Estimate biofluid-level read depth): Dataset-level loci were merged to create biofluid-specific exGLs; biofluid-level read depth being calculated as the maximum read depth of the merged

loci (**see Methods, Supplementary Figure S3**). We then ranked and binned biofluid-specific exGLs into abundance classes based on biofluid-level read depth. The ranking and binning were repeated for various subsets of exGLs (top 10, top 100, …, all exGLs) (**Figure 2A**). Within each subset, exGLs were sorted in descending order of read depth and divided into 10 equal-sized bins (abundance classes).

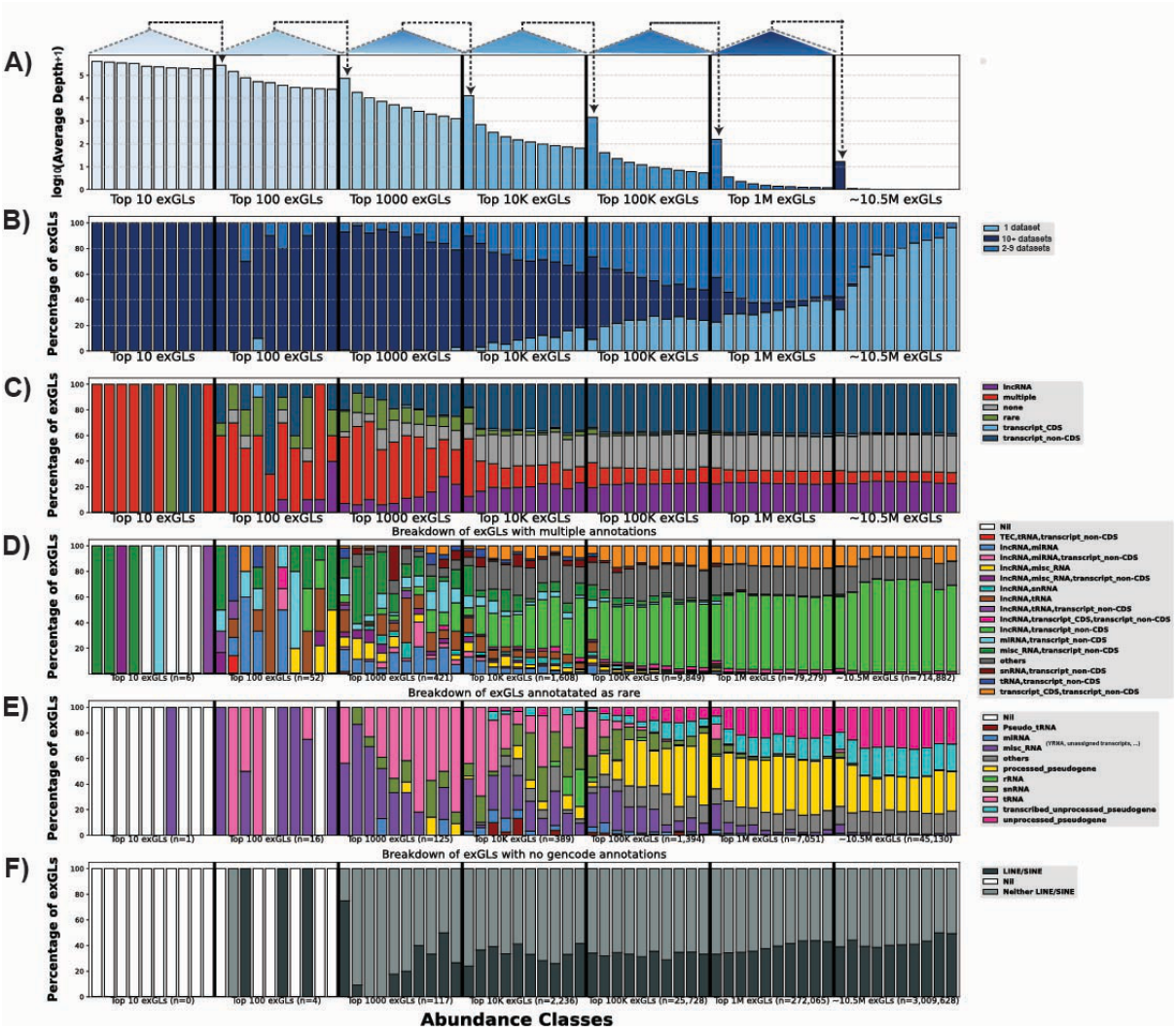


Figure 2: Plasma exGLs exhibit highly uneven abundance and diverse annotation profiles. *Plasma exGLs were divided into seven groups nested by abundance (left-most: 10 most abundant exGLs, right-most: all exGLs). (A) Log-transformed average read depth values for exGL within each abundance class. (B) Percentage of exGL in each abundance class mapped to loci found in 1 dataset, 2–9 datasets, or 10+ datasets.(C) Percentage of exGLs in each abundance class mapped to GENCODE RNA biotypes: lncRNA, protein-coding (subdivided into transcript_CDS and transcript_non-CDS), exGLs mapping to multiple annotations (labelled as "multiple"), exGLs mapping to any other RNA biotypes (labeled as "rare"), and exGLs not mapping to any GENCODE annotation (labelled as "none"). (D)*

Breakdown of "multiple" annotation exGLs, showing the most frequent annotations in each bar, with the remaining annotations grouped as "others". (E) Breakdown of "rare" annotation exGLs, showing the most frequent annotations, with the remaining annotations grouped as "others". misc_RNA includes Y_RNA, pseudogenes of Y_RNA, 7SL, and 7SK RNAs, and other transcripts that have not been assigned to any biotype. (F) Mapping of exGLs not overlapping any GENCODE annotation to LINE and SINE repeats. In panels (D)-(F), white bars indicate the absence of a given annotation in that abundance class, as inferred from panel (C).

Plasma exGLs showed a highly uneven read depth distribution, where a small subset of exGLs exhibit a much higher depth compared to the majority of plasma exGLs (**Figure 2A**). The top 10 exGLs exhibit read depth values ranging from 194,867 to 409,647. More abundant exGLs were more likely to be detected in multiple datasets (**Figure 2B**): Among the top 10,000 most highly expressed exGLs, 64% appeared in 10 or more datasets and 9% were present in only one dataset.

Section 3: Plasma exRNAs covering 2.2% of the genome have not previously been recognized as transcribed per GENCODE nor having a regulatory role per ENCODE

We next examined the distribution of exRNA biotypes in plasma by annotating exGLs with overlapping GENCODE v.47 transcripts (**Figure 2C**). The most abundant plasma exGL is observed at chr2:127798894-127799006 (GRCh38/hg38; read depth:409,647), and is annotated as both miscellaneous RNA (RNY4P7) and as non-coding portion of the transcript of a protein coding gene (WDR33, transcript non-CDS). Other top-10-ranked exGLs included lncRNAs (LINC00881, ENSG0000030284), miscellaneous RNAs (RNY4P10, RNY4, Y_RNA), miRNAs (MIR486-1, MIR486-2), tRNA (Gly_tRNA), and non-coding portions of transcripts of protein-coding genes (**Supplementary Table S4**).

Many exGLs across abundance classes mapped uniquely to loci annotated as non-coding portions of the transcripts of protein coding genes ("transcript non-CDS", ~38% of exGLs) and as "lncRNA" (~24% of exGLs). The third most frequent exGLs annotation category was "multiple" (~8% of exGLs), of which "transcript non-CDS" with "lncRNA" was the most frequent annotation combination (**Figure 2D**). The "rare" category (~0.5% of exGLs)

included “miRNA, “miscellaneous RNA”, “predicted tRNAs”, “processed pseudogene”, “unprocessed pseudogene” and “snRNA” among others (**Figure 2E**).

Among exGLs with no GENCODE annotations, ~42% mapped to LINE or SINE repeats (**Figure 2F**). ExGLs with no annotations were present across almost all depth groups and abundance classes.

Section 4: Star rating groups exGLs by levels of evidence and uniqueness of mapping

Although the broad genomic distribution of unannotated exGLs argues against these loci being purely mapping artifacts, it does not by itself establish equal confidence in the precise genomic origin of every locus. This distinction is especially important for repeat-overlapping exGLs, where similar sequences at multiple genomic locations could plausibly generate the same read. We therefore developed a tiered confidence framework that incorporates mapping stringency, repeat overlap, and reproducibility across datasets, assigning each exGL a star rating from zero to three stars. To build this rating, we augment the baseline exGL mapping (with up to one read basepair mismatch allowed) using zero-mismatch mapping, in which supporting reads were required to align to the reference genome with no mismatched bases. exGLs with less than 50% overlap with this zero-mismatch support received zero stars. Among exGLs with sufficient zero-mismatch support, loci overlapping repetitive elements were assigned one star. exGLs with zero-mismatch support and no repeat overlap received two stars if detected in a single dataset, and three stars if detected across multiple datasets.This rating revealed a similar pattern across all five biofluids. Over 90% of exGLs in CSF, Saliva, Serum, and Urine and ~55% in Plasma could be supported using zero mismatch mapping. Despite the lower percentage in Plasma, the higher-confidence tail was comparable across all five biofluids, with 19-25% of exGLs in each biofluid falling into the 2-3 star range **(Figure 3)**. Star rating was also weakly associated with locus length (**Supplementary Figure S4**). We anticipate that the star ratings attached to each exGL will empower users to choose subsets of exGLs that fit their analysis needs and to better interpret analysis results.

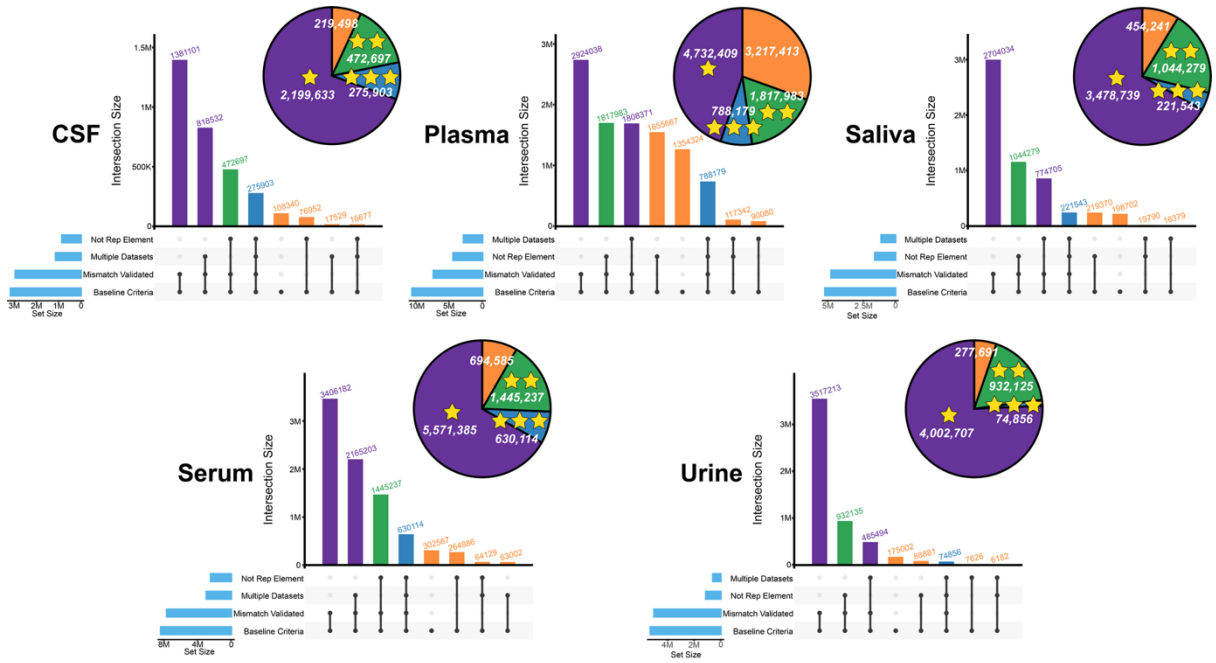


Figure 3. Distribution of exGLs across quality control and reproducibility criteria in each of the five biofluids. *Upset plots show the number of exGLs satisfying combinations of four criteria: Baseline Criteria (single read mismatch allowed), Mismatch Validated (validation using zero-mismatch read alignment), Multiple Datasets (reproducibility across independent datasets), Not Rep Element (exclusion of genomic regions annotated as repetitive elements). Horizontal bars show the total number of exGLs satisfying each criterion. Vertical bars, colored by star status (0-3 stars), indicate the number of exGLs satisfying specific criteria combinations. Pie charts summarize the fraction of exGLs for each star status (0-3 stars) within each biofluid.*

Having defined this confidence framework, we next asked whether the exGLs outside of GENCODE annotations had a lower star rating, suggesting potential false calls. We also considered low RNA abundance in the likely source tissue as a predictor of absence of GENCODE annotations covering exGLs. We focused on CSF exGLs and transcription levels in brain tissues because of the significant contribution of brain tissue RNA to the exRNA pool in CSF. As a source of brain tissue expression, we used GTEx brain RNA-seq signal tracks from the UCSC Genome Browser. CSF exGLs were classified as GENCODE-annotated or non-GENCODE and intersected with GTEx RNA-seq signal across 12 brain tissues, using one single high-quality healthy sample profile per tissue. Across the 12 GTEx brain tissues tested, 19.2–28.4% of GENCODE-annotated CSF exGLs overlapped GTEx brain RNA-seq signal, compared with only 3.8–7.2% of non-GENCODE CSF exGLs. This reduced intracellular representation was observed across star rating categories. Even among high-

confidence three-star exGLs, only 4.8–8.0% of GENCODE-unannotated CSF exGLs overlapped GTEx brain RNA-seq signal across tissues, compared with 40.1–46.1% of GENCODE-annotated three-star exGLs. Across tissues, both mean GTEx signal and GTEx overlap fraction were significantly lower for non-GENCODE exGLs by paired one-sided Wilcoxon signed-rank testing after Benjamini-Hochberg correction (adjusted p = 0.0019 for both comparisons). In an aggregated logistic regression model adjusting for brain tissue, CSF abundance decile, and star rating, non-GENCODE exGLs had significantly lower odds of GTEx expression than GENCODE-annotated exGLs (OR = 0.210, 95% CI = 0.209–0.211, $p < 2.2 \times 10^{-16}$). These results suggest that the CSF exGLs that fall outside of GENCODE annotations tend to be expressed at low levels in brain tissue irrespective of their star rating (**Supplementary Figure S5A-C**).

Section 5: exGLid identifiers and a public API/track hub make exGL annotations accessible as a searchable community resource

The study of exRNA biology and identification of exRNA biomarkers has so far been hampered by the lack of an exRNA naming system based on a community standard or reference database (Goh et al. 2022). To address this gap, we assign to each extracellular transcribed genomic locus (exGL) a unique identifier, exGLid (*see Methods*). Each exGLid is linked to a set of annotations (see *Methods*), including overlapping GENCODE gene names and biotypes, ENCODE regulatory elements, repeat elements, biofluid of origin, exRNA Atlas datasets where it was observed, and star rating. These annotations can be accessed via a user interface (https://exrna-atlas.org/exrna/exgl-query/) or programmatically through APIs (https://genboree.org/exrna/exgl-query/api/srvc*, see Methods*, **Supplementary Figure S6, Supplementary Tables S5-S7**). We also provide the top 10000 (filtered by read depth) exGL annotations in each biofluid as **Supplementary Table S8**. Additionally, all exGLs are viewable via UCSC Genome Browser track hub (https://genome.ucsc.edu/s/sreyya/exRNA%20loci). All smRNA-seq datasets and dataset-level bedgraphs and biofluid-specific exGL annotatin files used in this analysis are publicly

available through the exRNA atlas (https://exrna-atlas.org/ftps/exRNA-atlas/AtlasV3.0_data/).

Section 6: exGLs associate with higher GWAS SNP density and with changes in the density of basepairs under negative evolutionary selection.

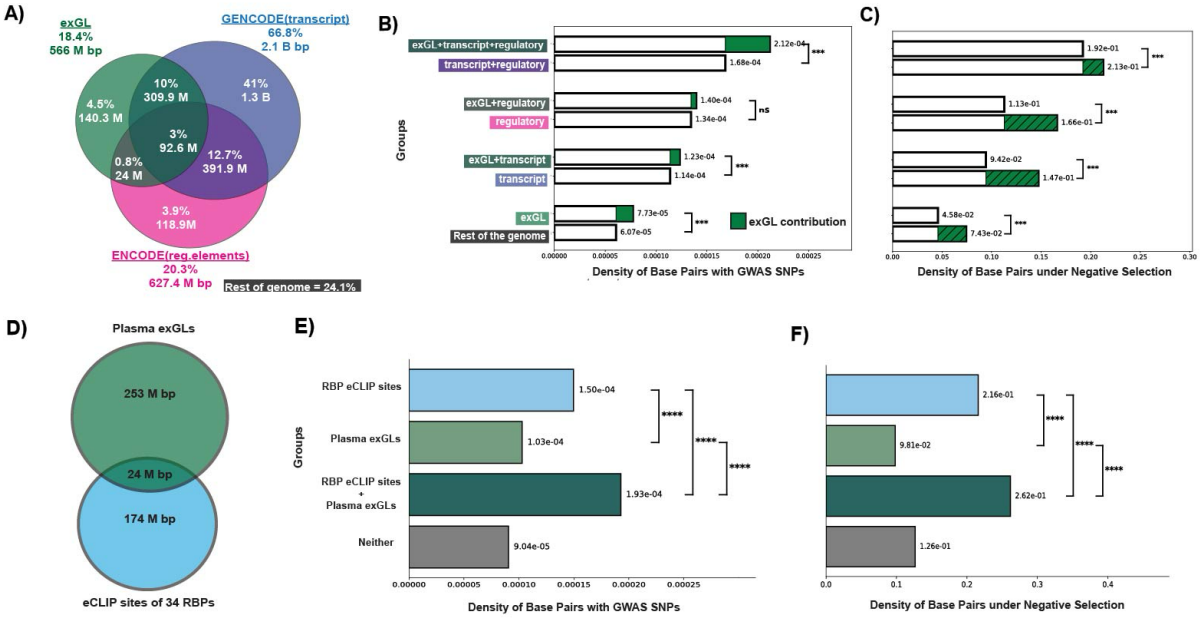


Figure 4. GWAS enrichment and negative selection associated with exGL loci. *(A) Genomic basepairs categorized into eight groups by three annotation types: exGL, GENCODE (transcripts), and ENCODE (regulatory elements), along with their intersections with the rest of the genome. Percentages indicate the proportion of genomic base pairs that fall into each category. (B) GWAS SNP density (GWAS SNPs per base pair) across the eight groups. Highlighted in green are GWAS SNP fractions attributable to exGL annotations. (C) Negative selection density (Number of bps with negative selection / total number of bp in category) for each category. Negative selection attributable to exGL annotations is shown (green-hashed). (D) Overlap of plasma exGLs and eCLIP binding sites from 34 exRNA-associated RBPs (LaPlante et al. 2023) categorized into three groups (E-F) Density of GWAS SNPs and negatively selected base pairs respectively across the three categories from D. Statistical significance was assessed using Poisson tests with multiple-testing correction. Panel C,F used 100,000 randomly selected bps in each category.*

To investigate the potential biological significance of exGLs from all biofluids, we explored association of exGL-associated basepairs with GWAS SNP density and with negative evolutionary selection during mammalian evolution. Toward this end, we assigned each base pair in the genome to exactly one of eight groups based on the intersections of the 3-way Venn diagram based on exGL, GENCODE (v.47) (transcript), and ENCODE (SCREEN regulatory elements) annotations (**Figure 4A**). Approximately 92.6 Mbp are shared across

all three annotation groups. Notably, a substantial portion of the genome (9.2%, 283.2 Mbp) is not transcribed per GENCODE annotations, and a subset of those (4.5%, 140.3 Mbp) lacked overlap with both GENCODE and ENCODE annotations. While these exRNAs have not previously been reported as even being transcribed, exRNAs arising from unannotated loci has been previously proposed based on their observations in plasma (Felden et al. 2022) and urine (Dogra et al. 2024).

We next examined the association of exGLs with the ~342K SNPs from the GWAS Catalog v1.0.2. We assigned each SNP to one of the eight groups defined by the three-way Venn diagram: exGL, transcript (GENCODE), and regulatory (ENCODE) annotations (**Figure 4B**). Loci overlapping exGL, transcript, and regulatory annotations displayed the strongest GWAS enrichment (2.12×10^{-4} GWAS SNPs per base pair), followed closely by "transcript + regulatory only" (1.68×10^{-4}) and "exGL + regulatory only" (1.40×10^{-4}) regions. This is consistent with the functional roles of transcribed and regulatory regions.

When compared to "rest of the genome", the "exGL only" annotations showed higher per-base density of GWAS SNPs. Furthermore, exGL annotations were more strongly associated with increased GWAS SNP density ("exGL contribution" indicated by green bars, **Figure 4B**), suggesting biological function. We next assessed evolutionary constraints using phyloP-derived measures of negative selection based on mammalian genome alignments (**Figure 4C**, Methods). Negative selection was estimated from 100,000 randomly sampled positions per category while preserving the underlying exGL length distribution. Categories overlapping transcript and regulatory annotations exhibited the strongest signatures of constraint. Surprisingly, exGL-containing regions showed a reduced density of negatively selected base pairs (**green-hashed bars, Figure 4C, Supplementary Table S9**), suggesting lack of evolutionarily conserved function across mammals.

In search of evidence of evolutionary conserved function within exGL loci, we next focused on the exGLs that overlap with eCLIP sites of RNA-binding protein (RBPs) (Van Nostrand et al. 2020). We tested this using eCLIP sites of 34 RBPs previously validated to bind to plasma exRNA (LaPlante et al. 2023) (**Figure 4E**). Plasma exGLs overlapping RBP eCLIP

sites showed the highest GWAS SNP density of the four categories tested (1.93 x 10^{-4} SNPs per base pair), exceeding RBP eCLIP sites alone (1.50 x 10^{-4}), plasma exGLs alone (1.03 x 10^{-4}), and regions that lacked either annotation (9.04 x 10^{-5}) (**Figure 4E**). The same exGL-eCLIP overlap category also showed the highest density of negatively selected base pairs (0.262), again exceeding RBP eCLIP sites alone (0.216), plasma exGLs alone (0.098), and either annotation (0.126) (**Figure 4F, Supplementary Table S10**). The excess of both evolutionary conservation and GWAS SNP density within exGLs overlapping binding sites of RBPs suggests a potential role for exGLs that overlap with RBP binding sites in human biology and disease etiology.

DISCUSSION

We provide the first map of extracellular Genomic Loci (exGLs) that give rise to exRNA transcripts in human plasma, serum, urine, CSF, and saliva. The exGL map leverages the human genome as a guide toward a comprehensive and unbiased analysis of molecular function relevant for human biology, similar to previous maps of protein coding exons (Harrow et al. 2012), transcribed non-coding RNAs (Djebali et al. 2012), and of regulatory elements (Dunham et al. 2012). A significant fraction of the map concerns fragments of longer transcripts, including binding sites of RNA Binding Proteins (RBPs) that may participate in the processing, loading, and transport of RNA into extracellular space (Villarroya-Beltri et al. 2013; Shurtleff et al. 2016). Another fraction of the map covers the genomic regions that have previously not been associated with any molecular function, possibly because they are expressed at low levels within cells. These exRNAs may have been previously overlooked due to a historical bias toward previously annotated regions of the genome. Thus, the exGL map opens a new frontier of unbiased exRNA research.

The exGL fraction of the genome is surprisingly large, of the same order of magnitude as the protein coding and gene regulatory fractions combined, with about 27 Mbps (>0.8% of the genome) within exGLs that are shared across all five biofluids, about 221 Mbps (>7% of genome) within exGLs that are shared by two, three, or four biofluids and > 11% of the

genome (349 Mbps) specifically detected only within one of the five biofluids. The fraction of basepairs not included in exGLs in any biofluid or included in most of the biofluids exceeds what is expected by chance, suggesting a level of concordance across biofluids. The extensive differences between biofluids may have various origins: some may be biological such as biofluid-specific cellular origin of exRNA; some may occur due to experimental confounding effects including differences in experimental parameters across biofluid-specific datasets. We anticipate that our map will provide a foundation for future studies that will shed more light on these differences.

The exGLs are numerous (~10.5M for plasma alone) and typically short, with a median size in the 22-28 bp size range. That size range is similar to the size of miRNA (Weber et al. 2010), the class of exRNAs that is highly abundant and most studied as a source of exRNA biomarkers (Mitchell et al. 2008; Condrat et al. 2020; Lucero et al. 2020; Thind and Wilson 2016) and for their role in post-transcriptional gene regulation (Valadi et al. 2007; Thind and Wilson 2016; O'Brien et al. 2018). Because extracellular miRNAs constitute a very minor fraction (0.02% in plasma) of genomic basepairs within the exGLs, our map exceeds the scope of currently studied exRNA biology by several orders of magnitude. However, it is also important to note that most exRNA Atlas datasets used for map construction utilized protocols that select for shorter RNA molecules, which may have led to under-representation of longer RNA species such as full-length mRNA and lncRNA transcripts. The abundance distribution of plasma exGLs reveals a long-tailed pattern, with a small subset of highly expressed exGLs consistently detected across multiple datasets, suggesting high reproducibility at even modest sequencing depths. In contrast, the majority of exGLs appear to be reproducibly detectable only at high sequencing depths. We anticipate that the exGL abundance information will help design sequencing experiments to meet the power and reproducibility requirements of future exRNA studies. The large range of exGL abundances suggest that sample-to-sample variation in sequencing depth may have significant effect on the ability to detect rare exGLs and quantify exGLs of medium abundance. Sequencing

depth should therefore be carefully considered during the design and analysis of profiling experiments.

The GWAS and evolutionary analyses presented here suggest that exGL annotations, especially those overlapping eCLIP binding sites of RBPs may capture biologically relevant genomic information. However, exGLs as a broad group do not show uniformly strong mammalian evolutionary constraint. This distinction is important: GWAS enrichment indicates trait-associated genomic relevance, but the constraint results suggest that additional context is needed to identify exGL subsets most likely to reflect conserved biological activity. Overlap with RNA-binding protein binding sites provides one such context (LaPlante et al. 2023). Plasma exGLs overlapping eCLIP sites of RBPs show stronger GWAS and negative-selection signals than exGLs or RBP-eCLIP sites alone, suggesting that this intersection identifies a more constrained subclass of extracellular RNA-associated loci. However, eCLIP binding sites do not by themselves establish that the same RBP–RNA interaction occurs in the extracellular space; they may also reflect intracellular RNA processing, stabilization, or localization events that precede extracellular release. So these analyses cannot distinguish whether the observed enrichment is driven by extracellular RNA function, intracellular post-transcriptional regulation, or both. Moreover, the GWAS analysis presented here does not identify trait-specific pathways or establish mechanisms through which extracellular RNA contributes to disease. Future studies will be required to identify specific biological processes and disease pathways.

We anticipate that the exGL identifiers (exGLids) will provide a reference system for exRNA studies. By annotating the exGLids with biofluid, dataset, GENCODE, ENCODE, repeat, abundance, and star ratings, the exGL Atlas provides a reference framework for reproducible exRNA analysis. The accompanying user interface, APIs, downloadable annotations, and UCSC Genome Browser track hub will allow researchers to query exGLs by genomic region, gene, annotation class, biofluid, or confidence tier.

Taken together, our findings suggest that extracellular RNA profiling reveals a large and previously under-characterized fraction of the human genome. The exGL map does not by

itself establish the function of each locus, but it provides a foundation for systematically prioritizing loci based on abundance, reproducibility, annotation context, and their star rating. While addressing only the most general questions about the exRNA fraction of the genome, we hope that the exGL map, analyses, and resources presented here will help catalyze future mechanistic and human studies into the specific roles of extracellular RNA communication in development, physiology, and human disease.

METHODS

ExRNA Atlas smallRNA-seq with ExceRpt on Cavatica

All small RNA sample fastq files from exRNA Atlas datasets in a major biofluid with >8 samples passing QC thresholds were uploaded to the Cavatica analysis platform. Dataset accession is provided in **Supplementary Table S1**. Fastq files were processed through the extra-cellular RNA processing toolkit (exceRpt) v5.0 (Rozowsky et al. 2019) pipeline. The exceRpt pipeline is a smallRNA-seq optimized workflow wrapper of the STAR RNA-seq alignment tool. Reads were preprocessed to remove barcodes, adapters, spike-in libraries and contaminants, then mapped to the hg38 reference assembly. Output bam files were converted to bedgraph format using BEDTools v2.30 (Quinlan and Hall 2010) (*bedtools genomecov*).

This data is hosted on the exRNA-Atlas in an FTP like structure and searchable by biosample and file name (https://exrna-atlas.org/ftps/exRNA-atlas/AtlasV3.0_data). For each biosample in a biofluid, users can access fastq.stats and related metadata.

exGL Creation and Read depth calculations.

To normalize counts across sample bedgraph files, the read counts at each locus were divided by the genome-mapping value (the number of sequencing reads mapping to the genome for each sample) and then multiplied by one million to generate normalized read depth per locus per sample.

Using BEDTools v2.30 (Quinlan and Hall 2010) (*bedtools unionbedg)*, we combined sample files within a dataset to create non-overlapping intervals across samples. We filtered out

intervals present in less than 2 samples. Then, using *bedtools merge*, adjacent loci of retained intervals from all samples were merged. Loci less than 18bp were then filtered out to yield our dataset-level loci set. Average read depth was calculated in two steps: (1) Using *bedtools map*, if multiple intervals merge into a single locus, their average depth is computed in each sample. (2) Once loci are merged across multiple samples within a dataset, the depth values from all samples are averaged.

For each biofluid, dataset-level filtered bedgraph files were combined to create biofluid-level exGLs. This involved merging overlapping or adjacent dataset-level loci using *bedtools merge*. We retain the largest read depth from the dataset-level loci merged to create each exGL.

Star rating Criteria

We assigned each exGL a star rating confidence annotation based on zero-mismatch validation, repetitive-element overlap, and reproducibility across datasets. Each exGL was first re-assessed under a stricter zero-mismatch alignment criterion. For each exGL, we evaluated overlap (atleast 50%) with exRNA loci produced from zero-mismatch alignment. exGLs with no zero-mismatch validation support, or with only the lowest validation category (NA or exactly 25%), were assigned a star rating of Zero.

Repetitive-element annotations were obtained from the RepeatMasker hg38 annotation (hg38.fa.out.gz; December 2013 assembly annotation, RepeatMasker open-4.0.5, Repeat Library 20140131), downloaded from the RepeatMasker database and converted to BED format. Validated exGLs overlapping a RepeatMasker element were assigned one star. Validated exGLs that did not overlap a repetitive element were assigned two stars when detected in a single dataset and three stars when detected in more than one dataset. Dataset reproducibility was determined from the contributing-dataset annotations generated for each exGL. For non-repetitive exGLs with zero mismatch support, loci detected in more than one dataset were assigned a star rating of Three, indicating higher confidence based on cross-dataset reproducibility. Remaining non-repetitive, zero mismatch supported exGLs detected in a single dataset were assigned a star rating of Two.

Overlapping base pairs across biofluids.

We analyzed all possible combinations of biofluids—CSF, serum, saliva, urine, and plasma—to quantify the base pairs uniquely associated with each combination. Overlapping intervals for each combination of biofluids were determined using *bedtools multiinter*, and total base pairs in each combination was calculated using custom script.

Annotating exGLs.

Extracellular genomic loci (exGLs) were annotated using GENCODE v.47 (Mudge et al. 2025) comprehensive gene annotation and GENCODE v.47 (Mudge et al. 2025) predicted tRNA genes (https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_47/gencode.v47.tRNAs.gtf.gz). GENCODE provides predicted tRNA annotations in which the gene_name field contains only numeric identifiers (e.g., "20658"), while the tRNA identity (e.g., "Asn_tRNA") is stored in the gene_type field. To make these annotations more interpretable, we reformatted the file so that the gene_name column included the corresponding tRNA name (e.g., "Asn_tRNA_20658"), and standardized the gene_type column to "tRNA" or "Pseudo_tRNA. Other annotations include regulatory regions from SCREEN v3 (Moore et al. 2020), and LINE/SINE repeat annotations retrieved from UCSC genome browser (Perez et al. 2025) (http://hgdownload.soe.ucsc.edu/goldenPath/hg38/bigZips/latest/hg38.fa.out.gz; accessed 02-09-2025). Annotation was performed using *bedtools intersect*, requiring at least 50% overlap of an exGL with an annotated loci or atleast 50% overlap of an annotated loci with an exGL. ExGLs overlapping protein-coding genes were further classified based on their specific genomic region: transcript_CDS for overlaps with coding sequence (CDS) regions, and transcript_non-CDS for overlaps with all other protein-coding regions including introns and UTRs. For classification within protein-coding regions, a minimum 1 bp overlap was permitted.

Abundance Classification of Plasma exGLs

Plasma exGLs were classified into abundance classes based on their read depth distribution using a custom R script. First, exGLs were ranked by depth and assigned to seven

cumulative groups, with each group expanding in size by a factor of 10 (e.g., top 10, top 100, etc.). Each group was then split into 10 abundance classes by sorting the exGL by depth (in descending order) and dividing the total number of exGL in the group into 10 equal-sized bins.

Positional concordance analysis

ExGL intervals for each biofluid were filtered. Gene coordinates were obtained from GENCODE v47. One thousand protein-coding genes were randomly selected, and *APOE* was also included. For each gene, every base pair was classified according to whether it was covered by exGLs in exactly 0-5 biofluids.

For each biofluid, the probability of exGL coverage was calculated as the fraction of the gene region covered by exGLs. Expected base-pair counts for each sharing category were calculated under an independence model using the biofluid-specific coverage probabilities. Enrichment was reported as:

log2 ((observed bp +1)/(expected bp +1)). The mean curve was calculated across the 1,000 randomly selected genes, with error bars representing the standard error of the mean.

For the RBP analysis, GRCh38-aligned ENCODE eCLIP narrowPeak files were downloaded from https://www.encodeproject.org/ for 34 RBPs previously validated to bind plasma exRNA. Multiple files for the same RBP were combined, and overlapping peaks were merged to generate one bedfile per RBP. Base pairs within each RBP peak set were classified by exGL coverage across exactly 0-5 biofluids, and expected counts were calculated using the same independence model with RBP-specific biofluid coverage probabilities.

ExGL abundance versus GTEx brain RNA-seq evidence

RNA-seq coverage BigWig files for 12 GTEx brain tissues were downloaded from the UCSC Genome Browser hg38 GTEx coverage directory (https://hgdownload.soe.ucsc.edu/gbdb/hg38/gtex/cov/). Each BigWig file was converted to bedGraph, and regions with RNA-seq signal greater than zero were retained and merged into contiguous intervals while recording mean and maximum signal. CSF exGLs were

intersected with these intervals using BEDTools, requiring at least 50% overlap of either the exGL or the GTEx interval. Overlap results were joined to the complete CSF exGL annotation, and exGLs were grouped into abundance deciles based on normalized read depth. GTEx overlap fractions and signal levels were compared between GENCODE-annotated and non-GENCODE exGLs across tissues and STAR-rating categories. Tissue-level differences were assessed using paired one-sided Wilcoxon signed-rank tests with Benjamini-Hochberg correction. An aggregated logistic regression model was used to test the association between GENCODE annotation status and GTEx overlap while adjusting for brain tissue, CSF abundance decile, and STAR rating.

GWAS Analysis

We identified intersections between exGLs, GENCODE v.47 transcripts (Mudge et al. 2025) (https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_47/gencode.v47.annotation.gtf.gz), and ENCODE regulatory elements (Moore et al. 2020) (https://downloads.wenglab.org/Registry-V4/GRCh38-cCREs.bed) using *bedtools intersect.* This classified base pairs into eight genomic groups, including one group outside the annotated regions. GWAS associations were downloaded from GWAS catalog v1.0.2 (Cerezo et al. 2025) (https://www.ebi.ac.uk/gwas/api/search/downloads/alternative). We filtered out traits influenced by interactions between two or more SNPs. After filtering, we had a total of 295,685 SNPs associated with various traits. We analyzed the distribution of these GWAS SNPs across our eight genomic groups identified by intersections between exGLs, GENCODE v47 transcripts, and ENCODE regulatory elements. *bedtools intersect* was used to map the SNPs to each genomic category.

To determine the GWAS SNP density for each of the eight genomic groups, we divided the number of SNPs in each category by the total number of base pairs in that category. Next, we determined the difference in SNP density (Δ Density) by comparing groups that contained a specific annotation (e.g., exGL) to their counterpart without that annotation (e.g., GENCODE only vs. exGL + GENCODE only). We applied the same approach to RBP-associated regions. ENCODE eCLIP sites from 34 RBPs previously validated to bind plasma

exRNA were merged and compared with plasma exGLs to define RBP-only regions, plasma exGL-only regions, regions overlapping both annotations, and regions containing neither annotation. GWAS SNP density was calculated for each category. Pairwise differences in density were assessed using two-sample Poisson rate-ratio tests with genomic base-pair coverage as the exposure, followed by Benjamini–Hochberg correction for multiple comparisons.

Evolutionary Analysis

We downloaded two sets of phylogenetic P scores from the PHAST package(http://compgen.cshl.edu/phast/), which were precomputed from alignments of 19 mammalian (http://hgdownload.cse.ucsc.edu/goldenPath/hg38/phyloP20way/hg38.phyloP20way.bw, accessed 11-18-2024) (Siepel et al. 2005; Pollard et al. 2010; Perez et al. 2025). 100,000 randomly selected positions were sampled from each of the eight genomic groups, and *bedtools intersect* was used to retrieve phyloP scores for each of the 100,000 positions. We used 100,000 positions to balance computational efficiency with statistical robustness. Because phyloP scores are calculated at single-nucleotide resolution, the full analysis would involve millions of positions. Positions were randomly sampled while preserving the exGL length distribution, providing a representative subset for estimating evolutionary constraint. A Gaussian Mixture Model with three components was applied to classify loci into negative, neutral, or positive selection groups based on phyloP score distributions. This data-driven approach allowed for an unbiased classification of evolutionary constraints across our eight groups. We then calculated the density of selected loci in each of the eight genomic groups by dividing the number of bps under negative selection by the total number of bps sampled in each group .The same workflow was applied to the RBP-associated categories defined above. One hundred thousand positions were sampled from the RBP-only, plasma exGL-only, RBP–exGL overlap, and neither categories. PhyloP scores were retrieved and classified using the same Gaussian mixture model, and negative-selection density was

calculated for each category. Pairwise differences were assessed using two-sample Poisson rate-ratio tests, followed by Benjamini–Hochberg correction.

exGLid Service

To facilitate standardized reporting and cross-study integration of extracellular RNA genomic loci, we established the GLid Registry and the exGLid Registry. Each exGL is assigned a unique exGLid, providing a stable reference linking genomic coordinates with associated annotations. The exGLid Registry currently contains approximately 30 million exGLids and serves as an internal reference system for generating and maintaining exGLids within the exRNA Atlas. To support data access and integration, exGL-associated annotations are available through the exGL Query API service. The exGL Query API service provides programmatic retrieval of exGL locations and associated metadata using exGLids, genomic coordinates, gene names and regulatory regions.

DATA ACCESS

All dataset-level bedgraphs used in this analysis are publicly available through the exRNA atlas (https://exrna-atlas.org/ftps/exRNA-atlas/AtlasV3.0_data/). Also included are biofluid-specific exGL annotation files. All scripts used for the computational analysis in this paper has been deposited at Zenodo (https://doi.org/10.5281/zenodo.21455434) and is publicly available.

COMPETING INTEREST STATEMENT

SD is a founder and owns equity for THryv Therapeutics and Switch Therapeutics, and consults for Thryv Therapeutics, none of which pertain to the work in this manuscript.

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